UK Heat Split: data sources and estimation methodology

Version 8 · Causeway Energies · August 2026

This revision supersedes the v7.1 methodology. Sections 4, 5, 6.1, 10.3, 10.5 and 12 are substantially rewritten; Sections 4.6, 5.4, 6.4 and 13 are new. A change log against v7.1 is given in Section 14.

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1. Conventions

1.1 Sourced quantities and author estimates

Every figure on the site is either traceable to a published source or is a Causeway Energies estimate. Estimates are marked with a dagger (†) wherever they appear, both in this document and on the site itself, and the contact address for challenge is published beside them. The distinction is maintained even where an estimate is well founded: the dagger denotes provenance, not doubt.

1.2 Purchased energy and delivered service

Two quantities are reported throughout and must not be conflated.

Purchased energy is what was bought: gas at the meter, electricity at the grid supply point, oil delivered. It is what appears on a bill.

Delivered service is what the building received: useful heat after combustion losses, useful cooling after the refrigeration cycle. For heat pumps delivered service exceeds purchased energy, because the difference is lifted from the environment rather than bought.

The energy-balance panel shows both, and the gap between them is the sum of combustion losses (a loss) and ambient heat harvested (a gain). These are opposite in sign and must not be netted into a single "efficiency".

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4. Cooling: the equipped fleet (observed)

Substantially revised at v8. The previous version described a latent-slope extrapolation and a saturation test that the data did not support; both have been withdrawn. See Section 14.

4.1 Data source and reconstruction

(unchanged from v7.1)

Cooling delivered by installed plant is inferred from national electricity demand. Half-hourly demand is taken from the National Energy System Operator (NESO) historic demand data and rolling update files [16]. Metered demand understates true underlying demand by the quantity supplied behind the meter, principally embedded solar photovoltaics, which is largest precisely on the hot, bright days when cooling load peaks. Underlying demand is therefore reconstructed by adding NESO estimates of embedded solar and wind generation to metered demand before analysis. The correction is material: embedded solar alone reaches of order 10–16 GW at midday in high summer against a metered demand of order 20–30 GW. Future-dated rows present in the update file, and days with other than 48 half-hourly values, are removed.

4.2 Within-class centring

(unchanged from v7.1, but see 4.5 for what it costs)

Summer electricity demand carries a strong non-cooling structure: it falls during the August holiday period and differs between weekdays and weekends. Binning raw demand against CDD conflates this structure with the cooling response and can yield a spurious negative uplift. To isolate the cooling signal, each daily underlying-demand value d_i is expressed as an anomaly about the mean of its (calendar-month × weekend/weekday) class:

$$a_i = d_i - \bar{d}(m_i, w_i)$$

where $\bar{d}(m, w)$ is the mean underlying demand over all days sharing month m and weekend/weekday status w. The operation is the within transformation of standard panel econometrics — a two-way fixed-effects demeaning on calendar month and weekend/weekday class [31] — applied here to isolate a weather signal rather than to estimate a panel coefficient. It is validated against synthetic data with a known injected slope (Section 4.4).

The term "demeaning" is avoided in public-facing copy in favour of "within-class centring", to prevent confusion with the ordinary meaning of the word.

4.3 Response curve: a continuous convex fit

Changed at v8. The response was previously binned by CDD and read as a step function, with a linear slope fitted to the low-CDD bins and extrapolated to define "latent" demand. Two faults were found.

The binning imposed a ceiling. The top bin held every day from CDD 4 to CDD 7.9 at a single value — fifteen days a year served the same figure whatever the weather. The response is now fitted continuously as

$$E(c) = k \cdot c^2$$

where c is cooling degree days on the daily mean of population-weighted outdoor air above the base in Section 2.3, and k is fitted by least squares through the observed bin means. The quadratic form is not assumed: it is what an undemeaned regression independently selects (Section 4.6), with an exponent of 2.0 to two significant figures and R^2 0.96 against the bin means. Removing the ceiling raises the annual total by about 26%. k is published in the payload as `k_quadratic` so the fit can be reproduced.

The response is convex, not saturating. The previous version tested for capacity saturation by extrapolating the low-CDD slope and looking for the observed curve to fall below it. The measured response does the opposite: it steps about +3 GWh per CDD across the cool bins against +17 across the warm ones. A linear extrapolation from the shallow end therefore sits *below* the observed curve at high CDD, and the difference it was meant to measure is structurally zero. The saturation tier has been withdrawn (Section 5.3), and the latent-slope line has been removed from the published chart.

4.4 Validation

(unchanged from v7.1)

The centring estimator was validated before deployment on synthetic data in which a known cooling slope was combined with an injected holiday depression of comparable magnitude to that seen in real August demand (Figure 3). Naive binning of the synthetic series reproduced the spurious negative-uplift artefact; within-class centring recovered the injected slope to within sampling error.

4.5 Why the observed and anchored figures differ

Rewritten at v8. The previous version reported an “order-of-magnitude discrepancy” between the observed response and the ECUK-anchored figure and left it unreconciled. The cause has since been identified and is a property of the estimator, not an error in either quantity.

Within-class centring removes the level along with the confound. Subtracting the (month × weekend-class) mean removes the mean itself. Cooling that runs steadily through July is contained in the July mean and is therefore invisible to the centred estimator; only day-to-day variation about that mean survives. The observed response is consequently a **lower bound by construction**, not an independent estimate of the total.

Quantitatively, the centred curve recovers about **15%** of the ECUK cooling anchor. Modelling the baseline explicitly instead — a constant, a day-type term, holiday terms, a heating limb and a convex cooling limb, fitted on **undemeaned** daily data — retains the intercept and recovers **74%** (Figure 5). Two estimators, one anchor, and the difference between them explained by the operation that separates them.

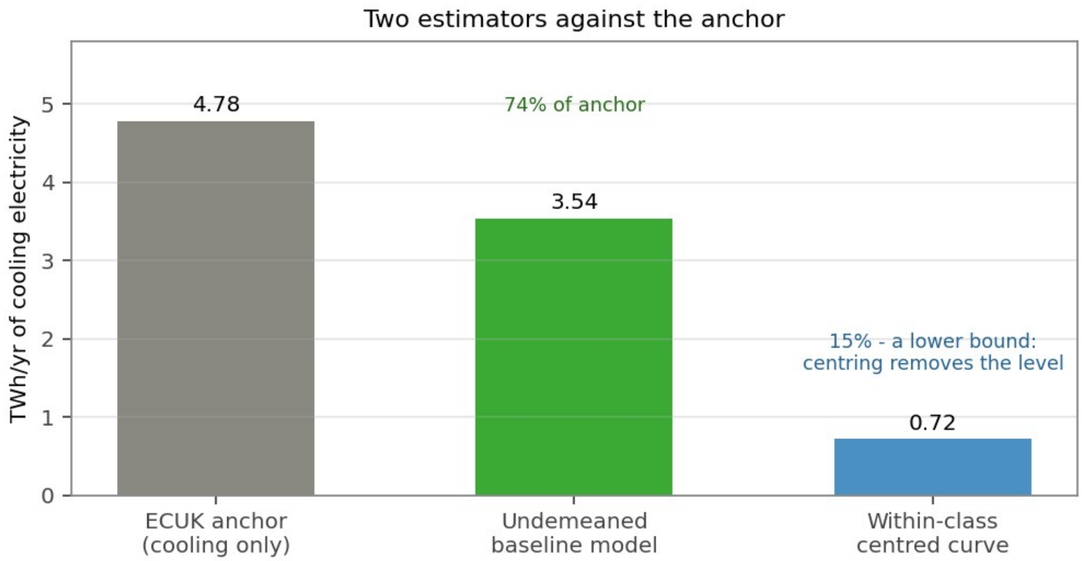

Figure 5

Figure 5. Two estimators against the ECUK cooling anchor. Within-class centring removes the level along with the confound and recovers about a seventh of it; modelling the baseline explicitly on undemeaned daily data retains the intercept and recovers 74 per cent.

This also resolves the tier arithmetic: the measured response and the anchored figure are not additive. The measured tier is a **subset** of the anchor, being the weather-varying part of the same equipped stock's consumption. The tiers that lie *outside* the anchor are those the anchor cannot see — buildings that want cooling and do not have it (Section 5), because a consumption survey cannot count a building that consumes nothing.

4.6 The undemeaned baseline model

New at v8. Because the centred estimator cannot recover a level, a second estimator is run offline for reconciliation. Daily GB demand, with embedded generation added back and no centring applied, is regressed on

$$D = \beta_0 + \beta_1 \cdot \mathrm{weekend} + \beta_2 \cdot \mathrm{HDD}(15.5) + \beta_3 \cdot \max(0, \bar{T} - T\_c)^2 + \Sigma\ \beta_j \cdot \mathrm{holiday}_j$$

where $\bar{T}$ is the population-weighted daily mean and T_c the cooling balance point, scanned over 11–20 °C. Because the baseline is modelled rather than removed, the intercept survives and T_c is identified.

On 400 days the best fit is T_c = **14.5 °C** with model R^2 0.769, giving an annual cooling electricity of **3.54 TWh** against the ECUK cooling limb of 4.78 TWh — 74%. The balance point is weakly identified: R^2 is nearly flat between 12 and 16 °C, and at 12 °C the fit returns 4.75 TWh, essentially the anchor. The plausible range of balance points therefore brackets the anchor.

Adding controls moves the cooling term by less than ±10%: holiday terms contribute the most (ΔR^2 +0.058 for the Christmas shutdown, +0.046 for bank holidays), a linear trend +0.018, and embedded solar +0.008. The solar coefficient enters with the wrong sign for a lighting effect and is collinear with temperature; it is not carried.

This estimator is **not** used in the published bill or carbon figures, which remain ECUK-anchored. It is a reconciliation diagnostic, and the case for switching the site onto it is deferred pending a full second cooling season.

4.7 Diurnal separation and the night response

New at v8. Two hourly diagnostics are computed and published, both from the retrospective store.

The diurnal slope profile. Summer demand is regressed on outdoor temperature separately for each hour of the day, centred within (year, month, day type, hour). Comfort cooling is a daytime load; process refrigeration runs continuously and its compressors still work harder as ambient rises. The profile separates them without requiring either level: an overnight floor of **0.261 GW/°C** at 04:00, rising to **0.676 GW/°C** at noon, on 3,048 summer hours (Figure 6). The floor is what a continuously running refrigeration fleet looks like; the daytime excess is comfort.

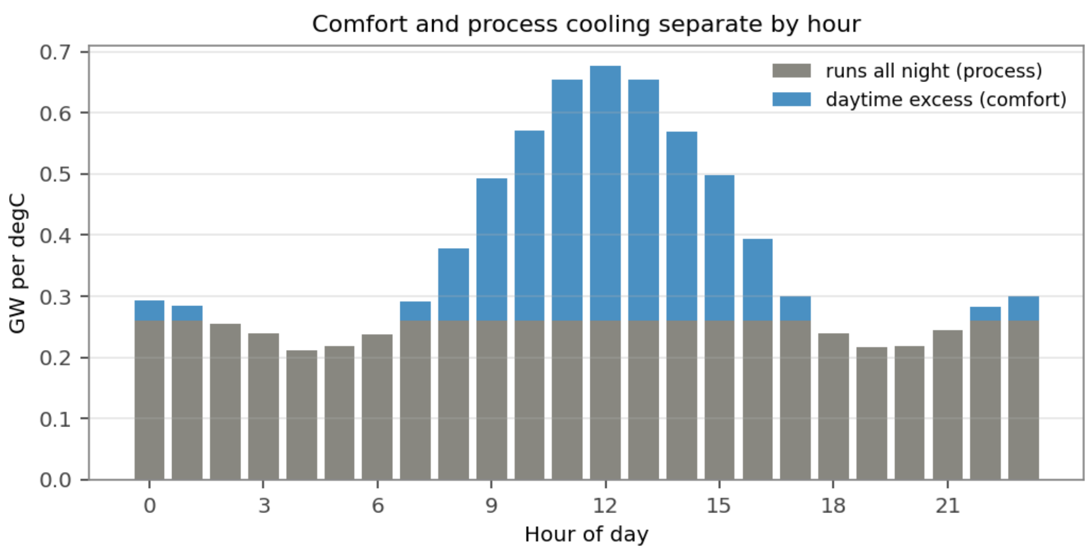

Figure 6

Figure 6. Summer demand regressed on temperature, hour by hour, centred within year, month, day type and hour. The overnight floor is cooling that runs regardless of the clock; the daytime excess above it is comfort.

The night elbow. Restricting to 01:00–05:00, May to September, and binning by temperature, the response is **not linear**: it is flat below about 15 °C and steep above (Figure 7). The break replicates independently in each summer on record — −0.074 then +0.300 GW/°C in 2025, −0.109 then +0.426 in 2026 — on 940 night hours.

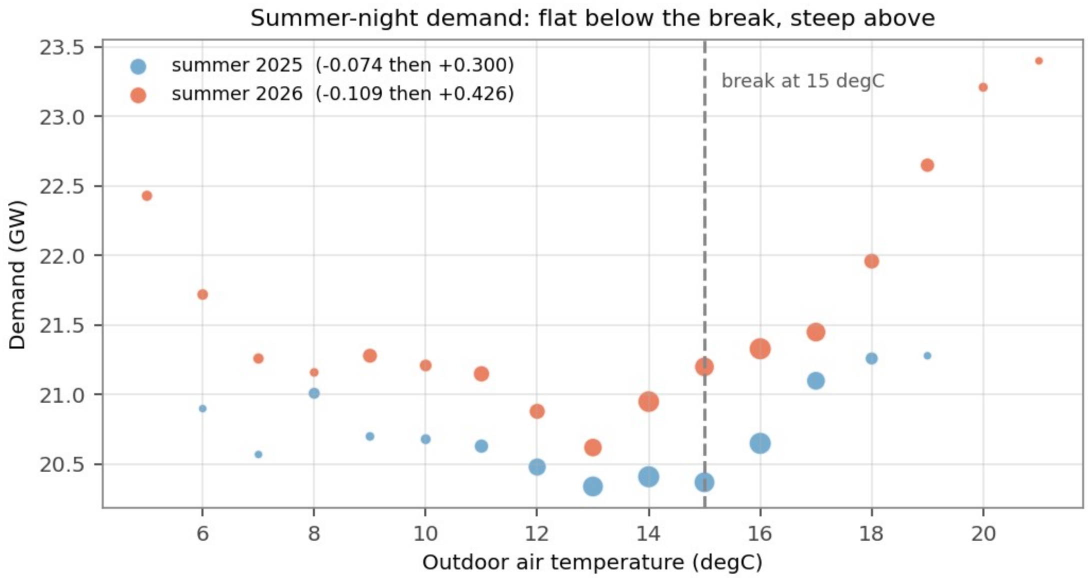

Figure 7

Figure 7. Summer-night demand against outdoor temperature, binned, by year. The response is flat below about 15 degC and steep above, and the break replicates independently in each summer on record. Point size shows hours per bin.

No level is inferred from either. An earlier attempt to invert the measured slope into an absolute refrigeration load via the COP identity was withdrawn: the flat segment below the break means the line being extrapolated does not exist, and the height of that floor is precisely the quantity aggregate demand cannot see. This is stated because the inverse method is superficially attractive and a reader may otherwise attempt it.

4.8 Tested and rejected

New at v8. Three candidate refinements were tested against the data and are recorded here as negative results.

Humidity. Latent load rises with absolute moisture, so moist-air enthalpy was a plausible explanation for the convexity in Section 4.3. Hourly dewpoint was fetched from the same ERA5 archive on identical population weights, converted to moist-air enthalpy against a 22 °C/50% RH reference (43.03 kJ/kg), and entered as an additional regressor. **Dewpoint adds 0.0014 to R^2 and enthalpy days add 0.0001**, against 0.046 for bank holidays. Decisively, the quadratic coefficient rose **+7%** when latent load would have made it fall. The convexity is therefore the **extensive margin** — each building is a step function and the fleet is the integral of those steps — not latent load. Humidity is not carried in the daily build.

Slope-to-level inversion. See Section 4.7.

The saturation tier. See Sections 4.3 and 5.3.

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5. Cooling: the comfort deficit (modelled)

Renumbered and rebased at v8.

5.1 Threshold and basis

Changed at v8. The previous version quantified overheating exposure as degree-hours above **26 °C**, citing the CIBSE operative-temperature criterion. This was a misapplication: **26 °C in CIBSE TM59 Criterion B is an INDOOR operative temperature** for bedrooms over 1% of night hours. It was applied to *outdoor* air, which is a different quantity and a much rarer event, and the tier was consequently understated by roughly five-fold.

The threshold is now **20 °C on the daily mean of population-weighted outdoor air**, derived from the 2017 Energy Follow-Up Survey indoor/outdoor regression, in which indoor bedroom temperature rises approximately 0.48 °C per 1 °C of outdoor temperature. Cooling degree-hours are computed as

$$CDH_{20} = \Sigma_d\, 24 \cdot \max(0, \bar{T}_d - 20)$$

This base is not the 15.5 °C cooling shape base of Section 2.3 and answers a different question. 15.5 °C is the balance point at which comfort cooling plant begins to run in buildings that have it; 20 °C is the outdoor daily mean above which buildings that have none become uncomfortable. Both appear on the site and the distinction is stated wherever they do.

A night-time urban heat island uplift is not in the numbers and would add an estimated 10–20% (†).

5.2 Stock and response

(largely unchanged; scenario values retained)

Three prevalence scenarios are carried and never averaged: a low case of 11% of dwellings from English Housing Survey self-reported overheating [9]; a central case of 25% (†); and a high case of over one-half, from the Climate Change Committee assessment of dwellings at overheating risk [10]. Non-domestic uncooled comfort space is taken as 190 Mm^2 (†) from BEES floor areas.

Thermal response is 0.2 kWh thermal per degree-hour per dwelling (†), equivalent to an effective cooled-zone UA of about 200 W/K, and 3.0 Wh thermal per m^2 per degree-hour for non-domestic space (†). These remain the dominant uncertainties.

Unit convention. This tier is a **thermal load computed directly from building physics and never divided by an efficiency**. Tier 1 by contrast is measured electricity multiplied *up* by the fleet EER. The two are reported in the same units and rest on entirely different assumptions; the site states this in the chart legend.

5.3 Tier structure

Changed at v8. The site previously published three tiers. The middle tier — "equipped but saturated", defined as latent demand minus delivered — has been **removed**, and the former tier 3 renumbered into its place.

The removed tier's premise was that the equipped fleet saturates at high load. The measured response steepens instead (Section 4.3), so `max(0, latent − delivered)` was structurally zero. Saturation would present as a *concave* response; the measured one is convex.

The computation is retained as a diagnostic and logged each run: a non-zero value would indicate the curve had turned concave, which is the evidence that would reinstate the tier.

The published tiers are therefore:

Tier	Content	Relation to the ECUK anchor
1	Weather-varying response of the equipped fleet	**Inside** it — about a seventh (Section 4.5)
2	Buildings with no cooling equipment at all	**Outside** it — a consumption survey cannot count non-consumption
0	Process cooling: data centres, cold stores, retail and food refrigeration	**Outside** ECUK U5, which is a services-buildings table. Of order 26 TWh/yr of service (†); named in the legend and deliberately not plotted, being several times the others combined

5.4 Delivery efficiency

Changed at v8. The comparison route efficiencies were previously stated as "an effective COP of order 20" for ground cooling and "a seasonal EER of order 3.5" for air-source, giving a six-to-one ratio.

The air-source figure is now **3.0, with a published range of 2.5–4.0**. The range matters: it is the largest single unverified assumption in the cooling chain, and it propagates one-for-one into every cooling service figure on the site. Legacy oversized plant runs nearer 2; modern commissioned VRF nearer 4.5. BEES excludes fan energy from its space-cooling line, so the boundary here is nearer chiller-plus-condenser than whole-system, which argues for the upper half of the range; UK part-load operation argues for the lower half.

What would resolve it: air-conditioning inspection reports under EPBD carry system type, age and rated efficiency for a large sample of UK non-domestic plant, and are the one dataset capable of producing a fleet-weighted seasonal figure rather than a judgement.

6.1 In-situ efficiencies, performance factors and the ECUK cooling split

Substantially revised at v8.

Combustion

Combustion appliances are derated by in-situ, not nameplate, efficiencies. From 13 August 2026 **space heating and hot water are converted at different efficiencies**:

Fuel	Space heating	Hot water	Basis
Gas	0.835	**0.73**	In-situ winter [5]; hot water derived below
Oil	0.82	**0.71**	As above
Bioenergy	0.70	0.70	Single figure — not covered by SAP Table 4b
Solid	0.55	0.55	As above

Why they differ. A boiler serving only a cylinder fires in short bursts, cools between them, and returns water above the condensing dewpoint — so it loses its latent recovery outright. SAP has carried separate winter and summer seasonal efficiencies for this reason since SAP 2009.

Derivation. The hot-water figures are the mean SAP 2012 Table 4b summer/winter ratio applied to the existing in-situ winter anchor, so the sourced winter values are untouched:

- gas: ratios 64/74, 74/84, 65/74, 75/84 → mean 0.879 × 0.835 = **0.73**
- oil: ratios 68/80, 72/84, 73/82 → mean 0.866 × 0.820 = **0.71**

Summer runs 82–89% of winter across the whole of Table 4b. Source: SAP 2012, Table 4b p207 [32].

The summer figure is applied to hot water in every month, not seasonally. SAP is explicit that η_water = η_summer for all twelve months: the distinction is between end uses, not between seasons.

Two caveats. Table 4b is SAP's *fallback* for boilers absent from the Product Characteristics Database, so it is conservative and skewed to older plant; a PCDB-weighted fleet figure would be better. And Table 4c deducts five points from both figures where a regular boiler has no interlock, which is not applied here.

Effect. The gas line acquires a seasonal shape it did not previously have. The effective gas efficiency, weighted across both end uses, now runs about 0.81 in January and **0.74 in July**, when the load is almost entirely hot water. It was a constant 0.835 in every month before. Useful annual hot water falls 66.2 → 58.8 TWh, so the space-plus-water total falls 1.8% and the electrification limit, which divides by that total, rises by about the same.

Heat pumps

Seasonal performance factors are 2.80 for air-source, from the Energy Systems Catapult Electrification of Heat demonstration [7]; 3.24 for ground-source, from Energy Systems Catapult in-situ monitoring of more than 1,700 installations [6]; and 5.0 for a networked geothermal ambient loop (†), for which no monitored UK fleet exists.

These annual factors anchor the model; they are not applied as constants in the hourly or daily engines. See Section 10.3.

The ECUK cooling and ventilation split

New at v8, and material. ECUK End Use Table U5 publishes services-sector **"cooling and ventilation" as a single electricity line**, held here at 10.4 TWh/yr. The site previously split this 50/50 between a weather-flat ventilation component and a weather-shaped cooling component. That split was undocumented and is now sourced.

The Building Energy Efficiency Survey (BEES, DECC/BEIS, fieldwork 2014–15), which is the survey ECUK's services end-use split is derived from, reports the components separately: **fans 6,030 GWh, space cooling 5,090 GWh and humidification 130 GWh** of electricity. That is **54% ventilation to 46% cooling**. US CBECS 2018, which separates the two natively, gives 18% of commercial electricity to ventilation against 14% to cooling — a 56:44 ratio from a considerably hotter climate, which corroborates the shape.

Ventilation has no coefficient of performance. Fans move air, not heat; there is no thermodynamic multiplication to apply, and fan work dissipates as heat inside the space, making it a cooling *load* rather than a cooling output. Ventilation is therefore reported as its own service at unity and is **excluded from the cooling figure**. This correction reduced published cooling delivered by approximately one third, from 20.8 to 14.35 TWh/yr on the annual anchor.

The cooling shape base

Changed at v8 from 18.0 °C to **15.5 °C** on the daily mean. 15.5 °C is the balance point at which comfort cooling begins in buildings with high internal gains, is the traditional UK degree-day base, and is what CIBSE TM46 applies across all sectors. It is corroborated by the undemeaned regression of Section 4.6, which places the best-fit balance point at 14.5 °C.

The Met Office cooling-degree-day base of 22 °C daily mean is a climate-comparison convention, not a building balance point, and would understate plant operation substantially. It is not used.

Degree days are fetched at bases 15.5, 18.0, 20.0 and 22.0. **CDD is convex in its base and the series cannot be interpolated between them** — averaging two bases overstates the middle.

6.4 The heat-price series

New at v8. A daily series of the commodity cost of delivered heat is published for four routes on two bases.

Wholesale. Gas is the National Gas System Average Price divided by the 0.835 boiler efficiency. The three electric routes are the daily electricity market index divided by **each route's own COP on that day's temperature** (Section 10.3), not by a fixed seasonal factor.

Retail. The same four routes at the Ofgem default-tariff cap in force on each date, which is a step function of policy decisions rather than a market series. The geothermal network route is priced at the **non-domestic** electricity rate, because a network operator buys on a commercial contract rather than the domestic cap. What is compared on that line is the operator's input cost, not a connected household's bill, which is a network tariff this model does not carry.

Three services, from v8.34. The chart carries a second toggle:

Service	Flow temperature	Gas efficiency
Space heating	Weather-compensated, 50 °C at −5 falling to 30 °C at +15	0.835
Hot water	**52 °C year-round** (†) — a cylinder needs its temperature whatever the weather	0.73
As delivered	Both, blended by the mix served that day	Both, blended

The blend is a harmonic mean weighted by delivered service, because what is being averaged is a cost per unit delivered:

```
effective COP = 1 / (w_space/COP_space + w_dhw/COP_dhw)
```

with w_space taken from the same HDD-shaped, summer-damped daily weighting the hourly engine uses (Section 10.2), and hot water flat. An arithmetic blend would flatter whichever service is dearer. **This is the only one of the three that moves with the load**, and it is where the separate hot-water efficiency becomes visible: it tracks space heating through the winter and hot water through the summer, when almost nothing else is being asked for.

The hot-water view sets an honest limit on the argument. Because the cylinder pins the flow temperature, none of the flow-reduction benefit of Section 10.9 is available to it and the lift never shrinks. The geothermal network's advantage over a gas boiler is about **1.7× in both January and July** on hot water, against 1.9× rising to 4.4× on space heating. The case for heat pumps is strongest exactly where the flow temperature can fall, and weakest on the load that pins it — which is the same mechanism as the single-header problem in non-domestic buildings.

Two electricity means are published per day: the flat mean, which the headline spark-gap ticker uses, and the **heat-weighted mean**, which is what a heat pump would actually pay because its load concentrates on the morning and evening peaks. The difference is the coincidence premium (Section 10.7), approximately 3% across a winter. The flat basis therefore understates every heat-pump route by about that much, and the site says so.

System Average Price is a balancing price — what National Gas pays to balance the system on the day. It tracks NBP day-ahead closely but is not the same instrument.

A caution on averaging. Any average of this series taken over *days* rather than over *heat delivered* will flatter whichever route performs best in mild weather. Unweighted, air-source shows a higher mean COP than ground source (4.88 against 4.06) and inverts the site's own SPF ordering; weighted by heat, the ordering is correct (3.49 against 3.66). On the thirty highest-heat days the figures are 2.21 and 3.08. Both weighted and unweighted values are published so the distinction is checkable.

10.3 Coefficient of performance and calibration

Substantially revised at v8. The equations are given in full because the flow-temperature results in Section 10.9 depend on them.

The Carnot form

For a heat pump lifting heat from a source at T_s to a delivery temperature T_f, both in kelvin, the thermodynamic ceiling is the Carnot coefficient of performance:

```
COP_Carnot = T_f / (T_f − T_s) = T_f / ΔT
```

Delivery temperature over lift. Real machines achieve a fraction of this:

```
COP = η · T_f / ΔT
```

with a floor of 8 K on ΔT (†), because the ratio diverges as the lift closes and no real machine does.

Route sources

Route	Source temperature
Air-source	Outdoor air less a 3 K evaporator approach, plus a defrost penalty of up to 12% between −5 and +7 °C
Ground source	**8.0 °C** — 11 °C ground less a 3 K brine approach
Geothermal network	**19.6 °C** (†) — a charged ambient loop, UTES or intermediate-doublet

The network source is 19.6 °C, not the 10 °C stated in v7.1. The correction is material: it is the temperature a *charged* loop reaches, and the difference between it and virgin ground is the basis of the seasonal-storage argument in Section 10.9.

Flow temperature

Flow is weather-compensated linearly between endpoints:

```
T_f(T_out) = 50 °C at T_out = −5 °C, falling to 30 °C at T_out = +15 °C,
             clamped outside that range (†)
```

Calibration of eta

η is **not assumed**. For each route it is solved so that the annual heat-weighted mean of the hourly COP reproduces that route's published seasonal performance factor. The current calibrated values are:

Route	η	Anchor SPF
Air-source	0.33198	2.80
Ground source	0.33646	3.24
Geothermal network	0.33436	5.00

That the three agree to within half a per cent is a check, not a coincidence. η is a statement about machine quality, which should not differ by route; what differs between routes is the source, and the Carnot term already carries it. The build refuses to publish if the three drift more than 15% apart.

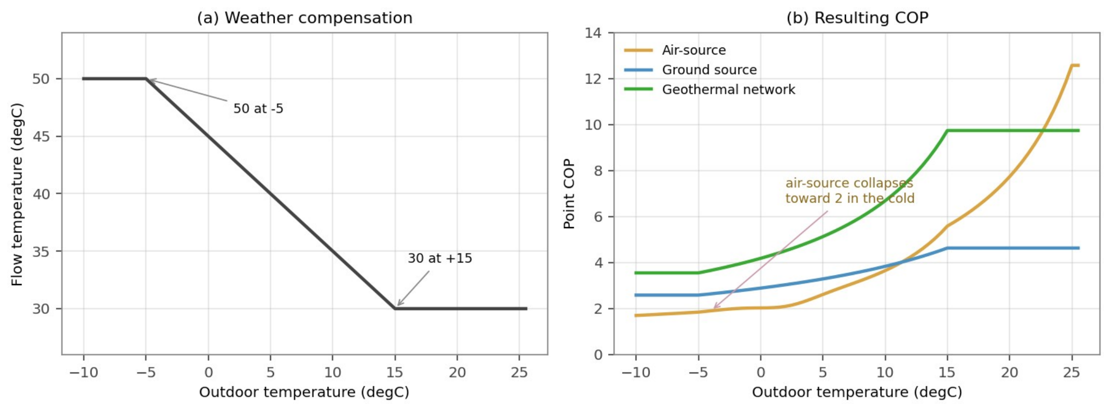

Figure 8

Figure 8. (a) The weather-compensation curve and (b) the point COP it produces for each route. Air-source falls toward 2 in the cold, exactly when heat is most wanted; a ground loop barely moves.

10.9 Sensitivity to flow temperature

New at v8.

Differentiating the Carnot form with respect to flow temperature, and noting that both numerator and denominator depend on T_f:

$$\partial COP/\partial T_f = \eta \cdot (-T_s) / \Delta T^2$$

The flow temperature cancels from the numerator; only the source survives. As a fractional change — which is what "per cent per kelvin" means:

$$(1/COP)\cdot\partial COP/\partial T_f = -T_s / (T_f \cdot \Delta T)$$

η cancels, so the percentage gain is independent of machine quality, and **ΔT remains in the denominator**, so the gain per kelvin is inversely proportional to the lift. Halving the lift doubles the value of each further kelvin (Figure 9).

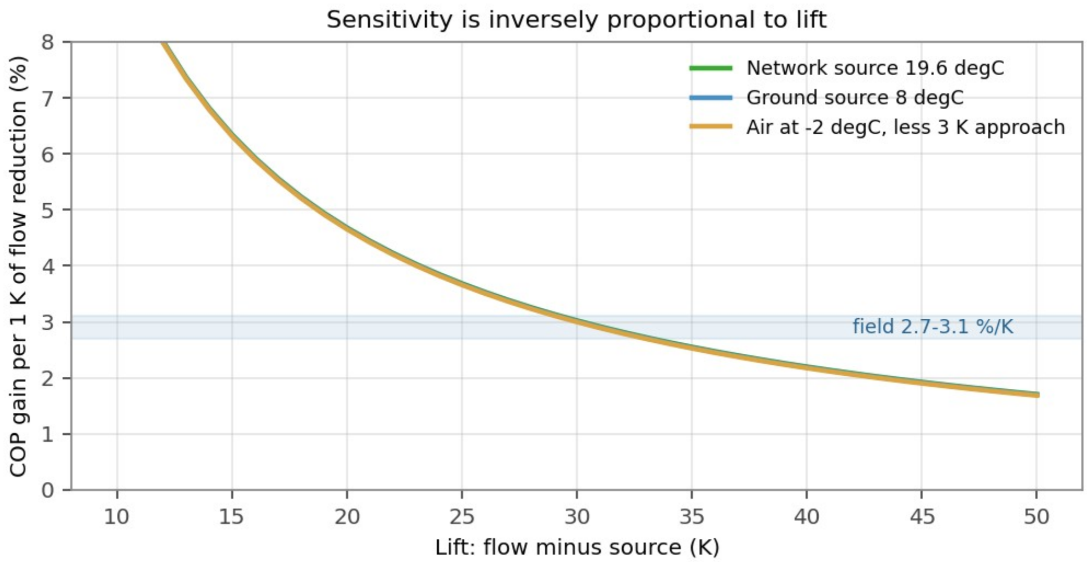

Figure 9

Figure 9. Fractional COP gain per kelvin of flow reduction, against lift. The route with the smallest lift gains most. The shaded band is the HeatpumpMonitor field range of 2.7 to 3.1 per cent per kelvin.

At a 45 °C flow: ground source at an 8 °C source has $\Delta T = 37$ K and gains 2.4% per kelvin; the network at 19.6 °C has $\Delta T = 25.4$ K and gains 3.6%. The ratio is simply 37/25.4.

Comparison with field data. HeatpumpMonitor.org publishes, from several hundred MID-metered UK systems at the SEPEMO H4 boundary, the regression

```
SPF_H4 = −0.1054 · (heat-weighted flow − outside) + 6.8762,   R² 0.547
```

This is the strongest correlation in that dataset. It implies **2.7–3.1% per kelvin** over a 25–35 K lift, against this model's 1.9% for air-source at the same condition (Figure 9). **The model therefore understates the effect on the one route for which field data exists.**

The most likely cause is that η is held constant. If the Carnot fraction itself improves as the lift shrinks — less superheat, better part-load matching — the field slope contains two effects where this model contains one. That would mean the network figure is conservative rather than optimistic. It is being put to OpenEnergyMonitor for comment and should be regarded as open.

The network figure of 3.6–4.2% per kelvin extrapolates beyond any monitored system and is daggered accordingly.

10.5 The capacity line

Changed at v8.

Requirements are set against a capacity line constructed as a de-rated block of dispatchable and interconnector capacity **plus the wind and solar actually generated in each hour**. Because outturn wind and solar are added hourly, the block must **exclude** de-rated wind, or wind is counted twice.

The block is derived identically for each winter from the NESO Winter Outlook:

```
block = ACS peak demand + de-rated margin − de-rated wind
```

Winter	ACS peak	Margin	Total de-rated	Less wind	**Block**
2024/25	59.8 GW	5.2	65.0	4.4	**60.6 GW**
2025/26	61.0 GW	6.1	67.1	~4.8 †	**62.3 GW**

The 2024/25 wind de-rating is NESO's own (30,532 MW installed de-rated to 4,416 MW). The 2025/26 figure is not published in the summaries and is scaled with installed capacity growth (†); the margin is sourced, the wind split is estimated.

Each winter is tested against its own fleet. Prior to v8 a single 62.0 GW block was applied to every hour, which was defensible while the store held one winter and became wrong the moment it held two.

One consequence for the binding-hour definition. With a ceiling that varies by winter, the binding hour is the hour of **least headroom**, not of highest absolute requirement. The two coincide when the store holds a single winter and need not otherwise. Both `_elec_limits` and `system_view` therefore maximise (requirement − block) and publish the block each figure was measured against.

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11. Automation, failure handling and reproducibility

(revised at v8: gates added)

A scheduled GitHub Actions workflow executes daily at 03:43 UTC. In addition to the four hourly gates (Section 10.8), three whole-build gates were added at v8 following defects that reached publication:

Annual-quantity plausibility. Trailing-twelve-month heating degree days must fall within 1,200–3,200. This gate exists because raising the regression window from 365 to 730 days caused every quantity that treated the window as a year to double — `hdd_trailing_12m` went 2,017 to 4,114, calibrated annual gas space heat 261.8 to 531.5 TWh, and the electrification limit halved from 98% to 49% — while all four test suites passed, because they check structure and internal consistency rather than whether an annual quantity spans a year. The window and the year are now separate constants (`WINDOW_DAYS` = 730, `ANNUAL_DAYS` = 365) and every trailing-twelve-month sum slices the latter.

Document structure. The published HTML is checked for content before the doctype, balanced script tags, and JavaScript leaking into markup. A failed string replacement once wrote 1,341 characters of JavaScript in front of the doctype, which rendered as visible text at the top of the live page while every syntax check passed: `node --check` on an extracted script proves the script parses, not that the document is well formed.

Scope. The build script's `main()` is walked for names read but never assigned in that scope and not module-level. A syntax check cannot see an out-of-scope name.

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12. Principal limitations

(revised at v8)

The following are the material limitations, restated for peer review.

— **The ECUK cooling and ventilation split** is now sourced (54:46, Section 6.1) but rests on a 2014–15 survey applied to later metered totals. No comprehensive UK non-domestic end-use survey has superseded BEES.

— **The cooling EER of 3.0** (range 2.5–4.0) propagates one-for-one into every cooling service figure and is the largest single unverified assumption in that chain (Section 5.4).

— **The comfort-deficit stock fractions and thermal-response coefficients** carry a several-fold range (Section 5.2).

— **The observed cooling response is a lower bound**, not an independent estimate, because within-class centring removes the level (Section 4.5). The undemeaned reconciliation recovers 74% of the anchor but is not yet used in the published figures.

— **The geothermal network SCOP of 5.0 and source of 19.6 °C** have no monitored UK fleet behind them. Every network figure on the site inherits this.

— **The flow-temperature sensitivity for the network route** extrapolates beyond any monitored system, and the model is known to understate the effect on the one route where field data exists (Section 10.9).

— **Geothermal output is not nationally metered**; the present-day bar remains the most weakly constrained figure on the site.

— **The capacity line is a static overlay**, now on **two** winters of observed weather rather than one, with per-winter blocks that must be revised each autumn. It cannot represent storage duration or network constraints.

— **Two winters is still a small sample.** The binding hour is now chosen across winters rather than describing the only one available, and 2025/26 holds it against 2024/25's own lower ceiling — but two is not a distribution.

— **Annual anchors lag** by roughly nine to fifteen months.

— **Hourly heat shape is modelled from weather, not metered.**

13. Corrections declared

New at v8. The site's practice is to declare corrections rather than silently restate. The following were found and fixed between v7.1 and v8, and are listed because a reviewer should be able to see what the process catches.

Defect	Effect	Resolution
Regression window used as if it were a year	Every annual quantity doubled; electrification limit halved 98% → 49% for one day	`ANNUAL_DAYS` separated from `WINDOW_DAYS`; plausibility gate added
Comfort-deficit threshold applied an indoor criterion to outdoor air	Tier understated roughly five-fold	Rebased to 20 °C daily mean from EFUS
Binned response curve imposed a ceiling	Hot days understated; annual total 26% low	Continuous quadratic fit
Saturation tier premise contradicted by the data	A published band that was structurally zero	Tier withdrawn; computation kept as a diagnostic
Ventilation credited with a coefficient of performance	Cooling service overstated by about one third	Ventilation reported separately at unity
Single capacity block applied to two winters	2024/25 tested against 2025/26's fleet	Per-winter blocks
Network source stated as 10 °C in v7.1 methodology	Understated the network COP and omitted the storage mechanism	Corrected to 19.6 °C, charged loop
One boiler efficiency applied to both end uses	Hot water credited with a condensing recovery it does not achieve; gas had no seasonal efficiency shape	Separate SAP-derived hot-water efficiencies, gas and oil

14. Change log against v7.1

Section	Change
4.3	Binned response replaced by a continuous quadratic; latent-slope extrapolation withdrawn
4.5	Observed-versus-anchored discrepancy explained rather than noted
4.6	**New** — undemeaned baseline model, balance point identification
4.7	**New** — diurnal separation and night elbow

Section	Change
4.8	**New** — humidity tested and rejected
5.1	Threshold rebased 26 °C indoor → 20 °C outdoor daily mean
5.3	Three tiers reduced to two; Tier 0 named
5.4	Cooling EER 3.5 → 3.0 with a published range
6.1	**New** — BEES 54:46 split; ventilation excluded from cooling; base 18.0 → 15.5 °C; **separate hot-water boiler efficiencies from SAP Table 4b**
6.4	**New** — heat-price series, both bases and three services
10.3	Carnot equations given in full; η calibration table published; network source 10 → 19.6 °C
10.5	Per-winter capacity blocks; binding hour redefined as least headroom
10.9	**New** — flow-temperature sensitivity and the field comparison
11	Three whole-build gates added
12	Limitations restated; one-winter caveat updated to two
13	**New** — corrections declared